Dynamic Dictionary Learning for Remote Sensing Image Segmentation

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Abstract

Remote sensing image segmentation faces persistent challenges in distinguishing morphologically similar categories and adapting to diverse scene variations. While existing methods rely on implicit representation learning paradigms, they often fail to dynamically adjust semantic embeddings according to contextual cues, leading to suboptimal performance in fine-grained scenarios such as cloud thickness differentiation. This work introduces a dynamic dictionary learning framework that explicitly models class ID embeddings through iterative refinement. The core contribution lies in a novel dictionary construction mechanism, where class-aware semantic embeddings are progressively updated via multi-stage alternating cross-attention querying between image features and dictionary embeddings. This process enables adaptive representation learning tailored to input-specific characteristics, effectively resolving ambiguities in intra-class heterogeneity and inter-class homogeneity. To further enhance discriminability, a contrastive constraint is applied to the dictionary space, ensuring compact intra-class distributions while maximizing inter-class separability. Extensive experiments across both coarse- and fine-grained datasets demonstrate consistent improvements over state-of-the-art methods, particularly in two online test benchmarks (LoveDA and UAVid). Code is available at https://github.com/XavierJiezou/D2LS.

1. Introduction

Remote sensing image segmentation aims to partition scenes into semantically consistent regions, supporting critical applications such as environmental monitoring [21], disaster assessment [18], and urban management [54]. Traditional methods often fail to address the inherent complexity of remote sensing data, including fine-grained structural variations [53] (e.g., thin and thick clouds), intra-class heterogeneity (e.g., varying cloud textures), and inter-class homogeneity (e.g., farmland and grassland) [20].

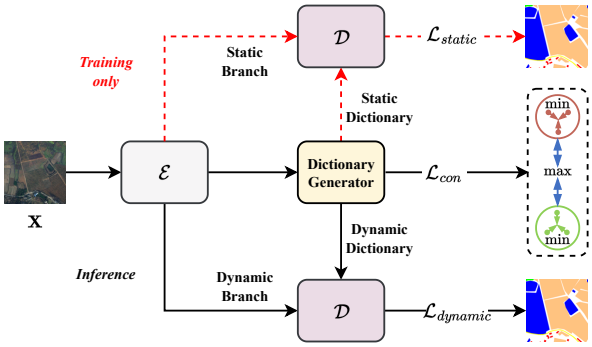


Figure 1. Overall pipeline of our dynamic dictionary learning. "$\mathcal{E}$" and "$\mathcal{D}$" denote the encoder and decoder, respectively.

Current research in remote sensing image segmentation has mainly explored three core directions. CNN-based methods like Hi-ResNet [6] and CDNet [19, 50] prioritize edge optimization and lightweight design through multi-scale architectures, while Transformer-based models such as DC-Swin [45] leverage global context via self-attention. Hybrid approaches (e.g., UNetFormer [46] and AerialFormer [20]) integrate CNN-Transformer strengths for efficient multiscale feature fusion, addressing challenges like cloud segmentation [12, 26] and domain adaptation [53].

Despite these developments, existing methods still face notable issues when confronted with high intra-class heterogeneity and inter-class similarity in remote sensing images. Specifically, most approaches rely on implicit representation learning, which often fails to adapt to subtle morphological or textural differences, such as differentiating thin vs. thick clouds [53] or distinguishing visually similar urban objects [20]. Furthermore, while attention mechanisms [2, 13, 22, 42] capture global context, they do

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not inherently ensure discriminative class-specific feature representations, leading to confusion among closely related categories. This limitation becomes more pronounced in fine-grained segmentation tasks [14, 53], where even minor appearance variations within a class can significantly affect model performance. These shortcomings highlight the need to encode class-aware embeddings explicitly.

To overcome these limitations, we propose a dynamic dictionary learning framework that explicitly constructs and refines class ID embeddings through iterative interactions between image features and semantic dictionaries. Our key innovations are as follows:

- We propose a dynamic dictionary learning framework that explicitly models class-aware semantic embeddings and leverages a contrastive loss to enhance intra-class compactness and inter-class separability, thereby improving performance, particularly in fine-grained remote sensing image segmentation tasks.
- We develop a modulator to convert the static dictionary into a dynamic one by generating attention maps from high-level semantic features, enabling input-specific adaptation and capturing subtle morphological variations within the same class.
- We design a multi-stage alternating cross-attention query decoder that progressively refines both image features and the semantic dictionary iteratively, facilitating mutual optimization and leading to more accurate and robust segmentation outcomes.

2. Related Work

2.1. Remote Sensing Image Segmentation

Remote sensing image segmentation methods can be categorized as coarse- and fine-grained approaches. Coarse-grained segmentation focuses on large-scale semantic understanding of scenes, such as urban landscapes and land-cover classification. Recent advances have integrated Transformers and convolutional neural networks (CNNs) to capture global context and local details. For example, UNetFormer [46] employs a lightweight ResNet encoder and a Transformer-based decoder for real-time urban scene segmentation, while Hi-ResNet [6] enhances edge details through multiscale branches and class-agnostic edge-aware loss. AerialFormer [20] combined hierarchical Transformers with multi-dilated CNNs to address challenges like object density and intra-class heterogeneity. Other studies, such as BANet [44] and ABCNet [29], further optimized feature fusion and computational efficiency through bilateral context modeling and lightweight architectures.

Fine-grained segmentation targets the precise delineation of objects such as thin and thick clouds. Cloud detection methods often leverage dual-branch architectures to balance spatial details and semantic context. DBNet [34] uses a Transformer-CNN hybrid with mutual guidance modules to refine cloud boundaries, and HRCloudNet [26] integrates hierarchical high-resolution features for complex cloud textures. MCDNet [12] addresses thin cloud detection via dual-perspective change-guided mechanisms. In contrast, lightweight models like SCNN [3] and CDNet series [19, 50] reduce computational costs using shallow networks or attention-based refinement. Recent domain adaptation frameworks, such as KTDA [53], further enhance generalization by aligning features across domains. These methods collectively advance fine-grained segmentation accuracy. However, most existing methods struggle to distinguish morphologically similar classes, resulting in ambiguous predictions for fine-grained scenarios. To address this, we propose a dynamic dictionary learning framework that explicitly models class-aware embeddings, thereby enabling a more robust discrimination of related categories.

2.2. Transformer and Attention Mechanisms

Transformer-based approaches such as DETR [2] reformulate object detection as a set prediction task, eliminating hand-crafted components like anchors and NMS. Subsequent works like DETR-style methods [9, 10, 36, 56] optimize query mechanisms and attention patterns to accelerate convergence and improve accuracy. For segmentation, MaskFormer [7, 8] unifies various segmentation tasks through mask classification, while SegViT [51, 52] and SegFace [37] leverage class tokens for semantic mask generation. These methods demonstrate the Transformer's versatility in dense prediction tasks.

Attention mechanisms [27, 28, 57, 58] play a pivotal role in enhancing feature discrimination and cross-modal interactions. Channel attention modules like SENet [22], CBAM [48], and ECANet [47] adaptively recalibrate channel-wise features, improving representation capacity. Cross-attention enables dynamic information exchange between heterogeneous data, as seen in CCNet [23] for criss-cross context aggregation. Such mechanisms also drive vision foundational models like Segment Anything [25], which utilizes prompt-based cross-attention for zero-shot segmentation. The synergy of diverse attention variants continues to push the boundaries of visual understanding systems. Inspired by DETR-style approaches and attention mechanisms, we explicitly model semantic labels with a learnable dictionary, further enabling progressive optimization via multi-stage alternating cross-attention, thereby effectively distinguishing morphologically similar classes.

3. Method

3.1. Overall Pipeline

The overall pipeline for our framework (see Fig. 1) can be divided into two key parts: training and inference.

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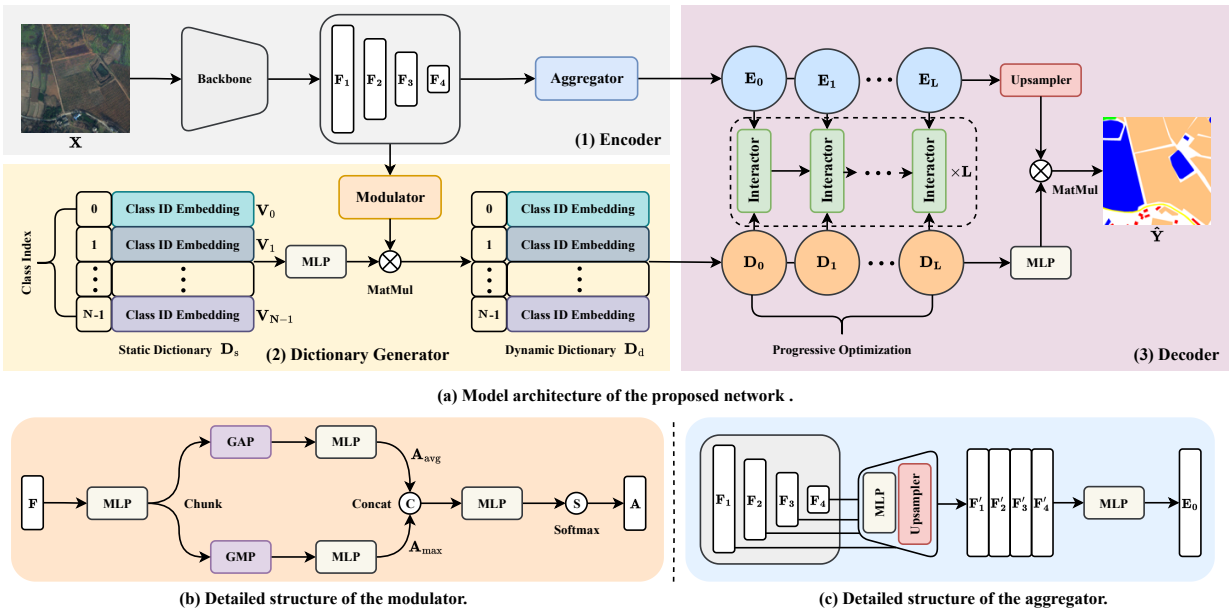


Figure 2. Overview of the proposed network architecture. Notably, we show only the dynamic branch for brevity.

3.1.1. Training

During training, the pipeline employs a dual-branch architecture: the encoder $\mathcal{E}$ processes the input image $\mathbf{X} \in \mathbb{R}^{3\times H\times W}$ and distributes the extracted features $\mathbf{F} \in \mathbb{R}^{C'\times H'\times W'}$ to two parallel branches: the static and dynamic branches. The static branch generates segmentation maps using a fixed static dictionary produced by a dictionary generator, while the dynamic branch operates with a continuously updated dynamic dictionary throughout training. Both branches share the same decoder. These processes can be formulated as:

$$\mathbf{F} = \mathcal{E}(\mathbf{X}), \quad \mathbf{O_s} = \mathcal{D}(\mathbf{D_s}, \mathbf{F}), \quad \mathbf{O_d} = \mathcal{D}(\mathbf{D_d}, \mathbf{F}), \quad (1)$$

where $\mathcal{E}(\cdot)$ and $\mathcal{D}(\cdot,\cdot)$ denote the encoder and decoder, respectively. $\mathbf{D_s} \in \mathbb{R}^{N\times C'}$ and $\mathbf{D_d} \in \mathbb{R}^{N\times C'}$ represent the static and dynamic dictionaries, while $\mathbf{O}_\text{s} \in \mathbb{R}^{N\times H\times W}$ and $\mathbf{O}_\text{d} \in \mathbb{R}^{N\times H\times W}$ are the segmentation outputs from the static and dynamic branches. $\mathbf{N}$ is the number of semantic categories, H, W, and C' denote the height, width of the image, and the number of feature channels, respectively.

In the training process, we achieve joint optimization by minimizing the loss functions of the static and dynamic branches, namely $\mathcal{L}_\text{static}$ and $\mathcal{L}_\text{dynamic}$. The total loss function is defined as a weighted sum of the individual losses:

$$\mathcal{L}_\text{total} = \lambda_\text{static} \cdot \mathcal{L}_\text{static} + \lambda_\text{dynamic} \cdot \mathcal{L}_\text{dynamic}, \quad (2)$$

where $\lambda_\text{static} = 0.4$ and $\lambda_\text{dynamic} = 1$ are the default hyperparameters. During training, the network is optimized by minimizing the total loss $\mathcal{L}_\text{total}$.

The loss for the static branch, $\mathcal{L}_\text{static}$, is composed of a weighted sum of the cross-entropy loss and the Dice loss, which is a common design in segmentation tasks [24, 37]:

$$\mathcal{L}_\text{static} = \mathcal{L}_\text{ce}(\mathbf{O}_\text{s}, \mathbf{Y}) + \mathcal{L}_\text{dice}(\mathbf{O}_\text{s}, \mathbf{Y}), \quad (3)$$

where $\mathbf{O}_\text{s}$ denotes the segmentation prediction from the static branch, and $\mathbf{Y} \in \mathbb{R}^{N\times H\times W}$ represents the ground truth segmentation label. The cross-entropy loss is employed for per-pixel classification. In contrast, the Dice loss focuses on the overlapping regions between the predicted and the ground truth segmentation maps, effectively mitigating the class imbalance problem, which is particularly important for segmentation tasks involving long-tail classes.

For the dynamic branch, the loss $\mathcal{L}_\text{dynamic}$ not only includes the same loss terms as the static branch but also incorporates a dictionary-based contrastive loss [39] to ensure the embedding representations for each class are unique and discriminative. Specifically, the dynamic loss is defined as:

$$\mathcal{L}_\text{dynamic} = \mathcal{L}_\text{ce}(\mathbf{O}_\text{d}, \mathbf{Y}) + \mathcal{L}_\text{dice}(\mathbf{O}_\text{d}, \mathbf{Y}) + \mathcal{L}_\text{con}(\mathbf{D}_\text{d}). \quad (4)$$

Moreover, the contrastive loss $\mathcal{L}_\text{con}$ consists of an intra-class compactness loss $\mathcal{L}_\text{intra}$ and an inter-class dispersion loss $\mathcal{L}_\text{inter}$:

$$\mathcal{L}_\text{con} = \frac{\mathcal{L}_\text{intra}}{\mathcal{L}_\text{inter} + \epsilon}, \quad (5)$$

where $\epsilon = 10^{-6}$ is used to prevent division by zero. The intra-class compactness loss measures the average Euclidean distance between the samples and their corresponding class centers:

$$\mathcal{L}_\text{intra} = \frac{1}{B\cdot N}\sum_{i=1}^{N}\sum_{b=1}^{B} \|\mathbf{D_d}^{(b,i)} - \boldsymbol{\mu}_i\|_2^2, \quad (6)$$

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where B denotes the batch size, N is the total number of classes, and $\mathbf{D}_{\mathrm{d}}^{(\mathbf{b},\mathbf{i})}$ represents the embedding of the b-th sample in class i. The class center $\boldsymbol{\mu}_i$ is computed as:

$$\boldsymbol{\mu}_i = \frac{1}{B}\sum_{b=1}^{B}\mathbf{D}_{\mathrm{d}}^{(b,i)}. \tag{7}$$

The inter-class dispersion loss aims to maximize the pairwise Euclidean distance between different class centers to enhance class separability:

$$\mathcal{L}_{\text{inter}} = \frac{2}{N(N-1)}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2^2. \tag{8}$$

The double summation iterates over all possible pairs of classes, ensuring adequate separation between them.

3.1.2. Inference

The network uses only the dynamic branch during inference, which utilizes the dynamic dictionary learned during training. The encoder processes the input image and generates feature maps, which are passed to the dynamic branch. The decoder then generates the segmentation maps. The dynamic dictionary complements the static dictionary by adaptively capturing the input-specific patterns, thereby improving the model's generalization to unseen data.

3.2. Proposed Network

The proposed network architecture (see Fig. 2) consists of an encoder, a dictionary generator, and a decoder, each playing a critical role in extracting features, learning dynamic dictionaries, and generating accurate segmentation maps.

3.2.1. Encoder

The encoder (as shown in Fig. 2a1) is responsible for extracting feature representations from the input image. Initially, the input image $\mathbf{X}$ is processed by a backbone, generating multi-scale feature maps $\{\mathbf{F}_i \in \mathbb{R}^{C_i\times H_i\times W_i}\}_{i=1}^{4}$, where $W_i = \frac{W}{2^i}$, $H_i = \frac{H}{2^i}$, and $C_i = C\times 2^i$. These feature maps are then passed through the aggregator, which performs feature mapping (Eq. (9)) and aggregation (Eq. (10)). The feature mapping step involves channel transformations, implemented by a multi-layer perceptron (MLP), and upsample interpolation to align all feature maps to the same dimensionality. After the mapping and interpolation, an MLP aggregates the features. The final output from the encoder is passed to the dictionary generator and the decoder.

$$\mathbf{F}'_i = \phi(\mathrm{MLP}(\mathbf{F}_i)), \quad i\in\{1,2,3,4\} \tag{9}$$

$$\mathbf{E}_0 = \mathrm{MLP}(\mathbf{F}'_1 \parallel \mathbf{F}'_2 \parallel \mathbf{F}'_3 \parallel \mathbf{F}'_4), \tag{10}$$

where $\{\mathbf{F'}_i \in \mathbb{R}^{C'\times H'\times W'}\}_{i=1}^{4}$ are the feature maps after feature mapping, and $H' = H_1, W' = W_1$. $\phi(\cdot)$ denotes the interpolation function. "$\parallel$" represents the concatenation operation along the channel dimension, and $\mathbf{E}_0 \in \mathbb{R}^{C'\times H'\times W'}$ is the final aggregated feature map.

3.2.2. Dictionary Generator

The dictionary generator is the core component of the proposed network and is responsible for creating static and dynamic dictionaries used in segmentation. These dictionaries help map feature information to class identifiers, enabling the model to distinguish different classes. The dynamic dictionary adapts during inference by modulating the static dictionary with input-specific attention. Unlike its static counterpart, it adjusts class ID embeddings to capture morphological variations across images, enhancing representation for nuanced inter-class distinctions.

Static Dictionary The static dictionary is a set of class ID embeddings used during the early stages of training. Each class ID embedding is associated with a learnable vector within the static dictionary. Once trained, these vectors are kept constant and do not change with input images, making them "static." As a result, the static dictionary remains unchanged during inference. The static dictionary can be represented as:

$$\mathbf{D}_{\mathrm{s}} = \{0:\mathbf{V}_0, 1:\mathbf{V}_1, \ldots, \mathbf{N}-1:\mathbf{V}_{\mathbf{N}-1}\}, \tag{11}$$

where $\mathbf{D}_{\mathrm{s}} \in \mathbb{R}^{N\times C'}$ denotes the static dictionary, class index serves as the key, and $\mathbf{V}_i \in \mathbb{R}^{C'}$ represents the embedding vector corresponding to the i-th class, which is the value in the dictionary. $\mathbf{N}$ is the number of classes.

Dynamic Dictionary The modulator (Fig. 2b) produces an attention map from encoder features to convert the static dictionary into a dynamic dictionary. The feature map $\mathbf{F} \in \mathbb{R}^{C_i\times H_i\times W_i}$ extracted from the encoder is first processed through an MLP layer. Specifically, we define $\mathbf{F} = \mathbf{F_4}$, which encapsulates high-level semantic information. Subsequently, $\mathbf{F}$ is split along the channel dimension into two separate chunks. One chunk undergoes global average pooling (GAP) followed by an MLP, while the other is processed through global max pooling (GMP) and a separate MLP. The outputs from these two branches are then concatenated along the channel dimension and fed into another MLP. Finally, the resulting feature representation is normalized using the softmax function, generating the attention map.

$$\mathbf{A}_{\text{avg}} = \mathrm{MLP}(\mathrm{GAP}(\mathbf{F}_{\text{avg}})), \mathbf{A}_{\text{max}} = \mathrm{MLP}(\mathrm{GMP}(\mathbf{F}_{\text{max}})), \tag{12}$$

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where $\mathbf{F}_{\text{avg}} \in \mathbb{R}^{\frac{C_i}{2} \times H_i \times W_i}$ and $\mathbf{F}_{\text{max}} \in \mathbb{R}^{\frac{C_i}{2} \times H_i \times W_i}$ represent the two chunks of the feature map $\mathbf{F}$. After concatenating the results of $\mathbf{A}_{\text{avg}} \in \mathbb{R}^{\frac{C_i}{2}}$ and $\mathbf{A}_{\text{max}} \in \mathbb{R}^{\frac{C_i}{2}}$, the final attention map $\mathbf{A}$ is computed as:

$$\mathbf{A} = \text{Softmax}(\text{MLP}(\mathbf{A}_{\text{avg}} \parallel \mathbf{A}_{\text{max}})), \tag{13}$$

where $\mathbf{A} \in \mathbb{R}^{N \times r}$ denotes the attention map, and $r = 4$ represents the reduced dimensionality. The static dictionary $\mathbf{D}_{\text{s}}$ is then transformed into the dynamic dictionary $\mathbf{D}_0 \in \mathbb{R}^{N \times C'}$ by matrix multiplication with the attention map $\mathbf{A}$:

$$\mathbf{D}_0 = \text{MLP}(\mathbf{D}_{\text{s}}) \otimes \mathbf{A}, \tag{14}$$

where $\mathbf{D}_0$ denotes the initial dynamic dictionary, $\otimes$ denotes matrix multiplication. The initial dynamic dictionary is then passed to the decoder for iterative interaction and updates.

3.2.3. Decoder

In the decoder, the dynamic dictionary $\mathbf{D}_0$ and the image feature map $\mathbf{E}_0$ interact over $\mathbf{L}$ stages for progressive optimization. During each stage, the dynamic dictionary and image feature alternately perform cross-attention queries, allowing for updates for both. The iterative process is:

$$\begin{aligned}
\mathbf{D}_l \in \mathbb{R}^{N \times C'} &= \text{softmax}\left(\frac{\mathbf{D}_{l-1}\mathbf{E}_{l-1}^{\top}}{\sqrt{d_k}}\right)\mathbf{E}_{l-1}, \quad \text{(D queries E)} \\
\mathbf{E}_l \in \mathbb{R}^{C' \times H' \times W'} &= \text{softmax}\left(\frac{\mathbf{E}_{l-1}\mathbf{D}_l^{\top}}{\sqrt{d_k}}\right)\mathbf{D}_l, \quad \text{(E queries D)}
\end{aligned} \tag{15}$$

where $l \in [1, \mathbf{L}]$ denotes the optimization stage. This alternating mechanism enables mutual enhancement between the dynamic dictionary and image features. Once the final feature map $\mathbf{E}_{\mathbf{L}}$ is obtained after $\mathbf{L}$ stages, the feature map is upsampled. Matrix multiplication is applied between the upsampled feature map and the dictionary $\mathbf{D}_{\mathbf{L}}$ to produce the final segmentation output:

$$\hat{\mathbf{Y}} = \phi(\mathbf{E}_{\mathbf{L}}) \otimes \text{MLP}(\mathbf{D}_{\mathbf{L}}), \tag{16}$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{N \times H \times W}$ represents the final output.

4. Experiments

4.1. Datasets

We evaluated our method on six remote sensing datasets for semantic segmentation, spanning a range of altitudes from UAV to satellite imagery. These datasets include coarse-grained (LoveDA [43], UAVid [35], Potsdam [15], Vaihingen [16]) and fine-grained (Cloud [14], Grass [53]) categories. For data splitting and preprocessing, the LoveDA, UAVid, Vaihingen, and Potsdam datasets follow the same protocol as in [24, 46]. In contrast, the Grass and Cloud datasets adopted preprocessing strategies identical to those used in [53]. LoveDA and UAVid are notably used in online test scenarios where ground truth labels are unavailable. For more details, please refer to **Appendix A**.

4.2. Implementation Details

Experimental Setup We set the size of the class ID embedding to 256 and the frequency of progressive optimization, denoted as $\mathbf{L}$, to 3. The backbone model was ConvNeXt-Base [31]. The training configuration utilized a batch size of 4, with the random seed fixed at 42 to ensure reproducibility. The optimizer was AdamW [33], with an initial learning rate set to 1×10^{-4} and weight decay set to 0.01. The learning rate was dynamically adjusted using a cosine annealing scheduler. All experiments were conducted on a workstation equipped with an NVIDIA A100.

Evaluation Metrics We followed the evaluation protocols of the comparative methods [20, 24]: Overall Accuracy (OA), intersection over union (IoU), F1 score, mean IoU (mIoU), and mean F1 (mF1). mIoU and mF1 are the averages of IoU and F1 scores across all classes. For readability, all metrics are presented in percentage form. Metric details and their formal definitions are provided in **Appendix B**.

4.3. Ablation Study

Effectiveness of Explicit Class Modeling As quantified in Tab. 1, our analysis reveals that implicit representation learning (by removing the dictionary generator) leads to consistent performance degradation (mIoU) across all datasets. The particularly pronounced impact on fine-grained segmentation tasks substantiates the effectiveness of explicit class modeling in the framework. Furthermore, additional ablation studies on the dimensionality of class ID embeddings and the backbone are available in **Appendix C**.

Table 1. Effectiveness of explicit class modeling.

Dictionary Generator	mIoU ↑		
	LoveDA	Vaihingen	Grass
×	54.81	84.40	50.70
✓	**55.27**	**85.27**	**51.96**

Static Branch vs. Dynamic Branch Tab. 2 revealed the complementary roles of the loss function in different branches. Using only static loss (static branch) yields limited performance, while adding dynamic loss (dynamic branch) improves segmentation performance. The contrastive loss provides critical discrimination power - its inclusion boosts Grass performance by +0.43, with LoveDA/Vaihingen also peaking at 55.27/85.27. These results confirm that the dynamic branch is more effective and that jointly optimizing both branches with contrastive loss consistently yields the best performance.

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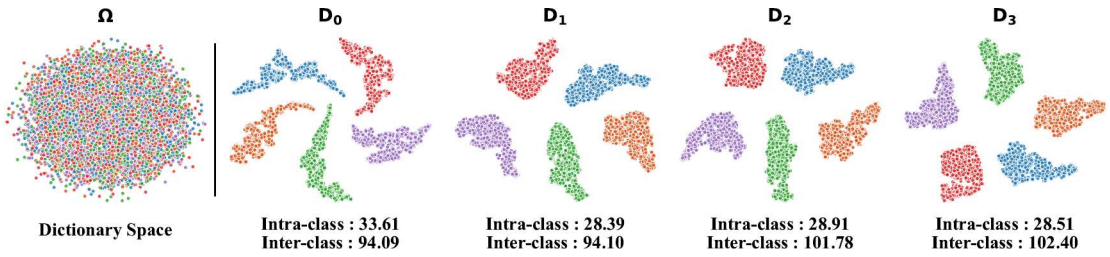


Figure 3. t-SNE visualization of dictionary distributions across different interaction stages.

Table 2. Ablation study on loss function of different branches.

Method	mIoU ↑		
	LoveDA	Vaihingen	Grass
$\mathcal{L}_{\text{static}}$	54.80	85.10	50.96
$\mathcal{L}_{\text{dynamic}}$	54.93	85.17	51.38
$\mathcal{L}_{\text{static}}$ + $\mathcal{L}_{\text{dynamic}}$ (w/o $\mathcal{L}_{\text{con}}$)	54.95	85.17	51.53
$\mathcal{L}_{\text{static}}$ + $\mathcal{L}_{\text{dynamic}}$ (w/ $\mathcal{L}_{\text{con}}$)	**55.27**	**85.27**	**51.96**

Table 3. Performance comparison across interaction stages.

L	mIoU ↑				mF1 ↑	
	LoveDA	UAVid	Grass	Cloud	Potsdam	Vaihingen
1	54.84	68.79	51.05	83.09	93.29	91.83
2	55.00	69.49	51.21	82.95	94.09	**91.88**
3	**55.27**	**70.90**	**51.50**	**83.10**	**94.70**	91.76
4	54.84	67.88	50.90	80.97	93.87	91.85

Interaction Stages The feature-dictionary interaction stages progressively update class-aware semantic embeddings through multi-stage alternating cross-attention between image features and the dictionary. Experiments in Tab. 3 showed dataset-specific preferences: $\mathbf{L} = 3$ yields optimal performance on most datasets, including LoveDA, UAVid, Grass, and Cloud, while $\mathbf{L} = 2$ suffices for Vaihingen (91.88 mF1). Excessive stages ($\mathbf{L} = 4$) reduce segmentation performance on the Cloud dataset, indicating overfitting risks. Visualization of t-SNE (Fig. 3) on the Grass dataset quantifies dictionary evolution: intra-class distances tighten ($\mathbf{D}_0$: 33.61 → $\mathbf{D}_3$: 28.51), while inter-class distances increase. We selected $\mathbf{L} = 3$ as the default, balancing model performance and computational efficiency.

Effectiveness of Each Component We systematically validated the necessity of each component by disabling individual components. As shown in Tab. 4, removing the modulator resulted in the most substantial decline in segmentation performance on the Vaihingen dataset, highlighting its critical role in handling complex urban scenes. Disabling the aggregator leads to the most significant performance drop on the Grass dataset, indicating its importance for fine-grained feature integration. The interactor contributed consistently across all datasets, demonstrating its capacity to refine ambiguous boundaries. The full model achieved peak performance, proving that these modules provide complementary functionalities rather than isolated improvements. This comprehensive ablation validates that all components are essential for segmentation performance.

Table 4. Effectiveness of each component. Mod., Agg., and Int. denote modulator, aggregator, and interaction, respectively.

Mod.	Agg.	Int.	mIoU ↑		
			LoveDA	Vaihingen	Grass
×	✓	✓	54.95	85.17	51.35
✓	×	✓	54.66	82.54	48.52
✓	✓	×	54.55	85.14	50.95
✓	✓	✓	**55.27**	**85.27**	**51.96**

4.4. Comparison with SOTA Methods

Quantitative Analysis Our method achieved consistent SOTA performance on six remote sensing semantic segmentation benchmarks. It also balances efficiency and performance well, with detailed computational efficiency comparisons (parameter count and FLOPs) in **Appendix D**.

Coarse-grained Segmentation Our method achieved state-of-the-art performance across all coarse-grained datasets. On the LoveDA dataset (Tab. 5), we surpassed SFA-Net by +0.4% mIoU, particularly in challenging categories like agriculture and building. The UAVid results (Tab. 6) further validated our superiority in dynamic urban scenes, achieving 70.9 mIoU by learning distinct representations for different categories. For high-resolution aerial datasets, our approach established new benchmarks on Potsdam and Vaihingen (Tab. 7), with dominant performance in large-scale segmentation scenes.

Fine-grained Segmentation As shown in Tab. 8, our method demonstrates remarkable advantages in distinguishing subtle class variations. On the Grass dataset, we achieve 66.27 mF1, with significant gains in differentiating coverage grades. For the Cloud dataset, our approach establishes an absolute lead, benefiting from the dynamic dictionary's adaptability to image feature variations. The 89.65 mF1 on

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Table 5. Comparison of semantic segmentation performance on the LoveDA dataset.

Method	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU ↑
TransUNet [4]	43.0	56.1	53.7	78.0	9.3	44.9	56.9	48.9
DC-Swin [45]	41.3	54.5	56.2	78.1	14.5	47.2	62.4	50.6
UNetFormer [46]	44.7	58.8	54.9	79.6	20.1	46.0	62.5	52.4
Hi-Resnet [6]	46.7	58.3	55.9	80.1	17.0	46.7	62.7	52.5
AerialFormer [20]	47.8	60.7	**59.3**	81.5	17.9	47.9	64.0	54.1
SFA-Net [24]	**48.4**	60.3	59.1	**81.9**	**24.1**	46.2	64.0	54.9
Ours	47.6	**61.2**	59.1	81.6	23.8	**48.8**	**64.8**	**55.3**

Table 6. Comparison of semantic segmentation performance on the UAVid dataset.

Method	Clutter	Building	Road	Tree	Vegetaion	Moving Car	Static Car	Human	mIoU ↑
DANet [17]	64.9	58.9	77.9	68.3	61.5	59.6	47.4	9.1	60.6
ABCNet [29]	67.4	86.4	81.2	79.9	63.1	69.8	48.4	13.9	63.8
BANet [44]	66.7	85.4	80.7	78.9	62.1	69.3	52.8	21.0	64.6
SegFormer [49]	66.6	86.3	80.1	79.6	62.3	72.5	52.5	28.5	66.0
UNetFormer [46]	68.4	87.4	81.5	80.2	63.5	73.6	56.4	31.0	67.8
SFA-Net [24]	70.2	89.0	82.7	80.8	64.6	**77.5**	**67.5**	30.7	70.4
Ours	**71.0**	**89.7**	**83.2**	**82.1**	**66.1**	75.0	59.0	**41.4**	**70.9**

Table 7. Comparison of semantic segmentation performance on the Potsdam and Vaihingen datasets.

Method	mF1 ↑		
	Potsdam	Vaihingen	Average
DANet [17]	88.9	79.6	84.3
ABCNet [29]	92.7	89.5	91.1
Segmenter [41]	89.2	84.1	86.7
BANet [44]	92.5	89.6	91.1
SwinUperNet [30]	92.2	89.8	91.0
DC-Swin [45]	93.3	90.7	92.0
UNetFormer [46]	93.5	90.4	92.0
AerialFormer [20]	94.1	90.1	92.1
SFA-Net [24]	93.5	91.2	92.4
Ours	**94.7**	**91.9**	**93.3**

Cloud highlights our capacity to resolve ambiguous boundaries through dynamic dictionary learning. These results collectively confirm that our dynamic dictionary learning paradigm adapts effectively to large-scale scene variations and fine-grained semantic distinctions.

Visualization Results As shown in Fig. 4, our method accurately segments dome-shaped buildings in LoveDA, captures tiny human targets in UAVid without interference from vegetation and clutter, and correctly identifies rough road surfaces in Potsdam despite ground truth mislabeling. Additionally, it enables precise cloud-type separation in Cloud and enhances vegetation classification in Grass (Fig. 5). More visualization results can be found in **Appendix E**.

Table 8. Comparison of semantic segmentation performance on the two fine-grained datasets.

(a) Grass

Method	mF1 ↑
FCN [32]	61.99
PSPNet [55]	62.55
DeepLabV3+ [5]	62.50
UNet [40]	62.34
SegFormer [49]	62.82
Mask2Former [8]	58.91
DINOv2 [38]	61.70
KTDA [53]	65.01
SFA-Net [24]	65.76
Ours	**66.27**

(b) Cloud

Method	mF1 ↑
MCDNet [12]	42.76
SCNN [3]	52.41
CDNetv1 [50]	45.80
KappaMask [11]	68.47
UNetMobv2 [1]	56.91
CDNetv2 [19]	70.33
HRCloudNet [26]	71.36
KTDA [53]	60.08
SFA-Net [24]	84.64
Ours	**89.65**

5. Conclusion

In this paper, we propose a dynamic dictionary learning framework to address the limitations of distinguishing similar categories in complex scenes caused by implicit representation learning in remote sensing image segmentation. Our method adaptively models morphological variations while preserving semantic discriminability by iteratively refining class-aware ID embeddings through alternating cross-attention querying and contrastive constraints.

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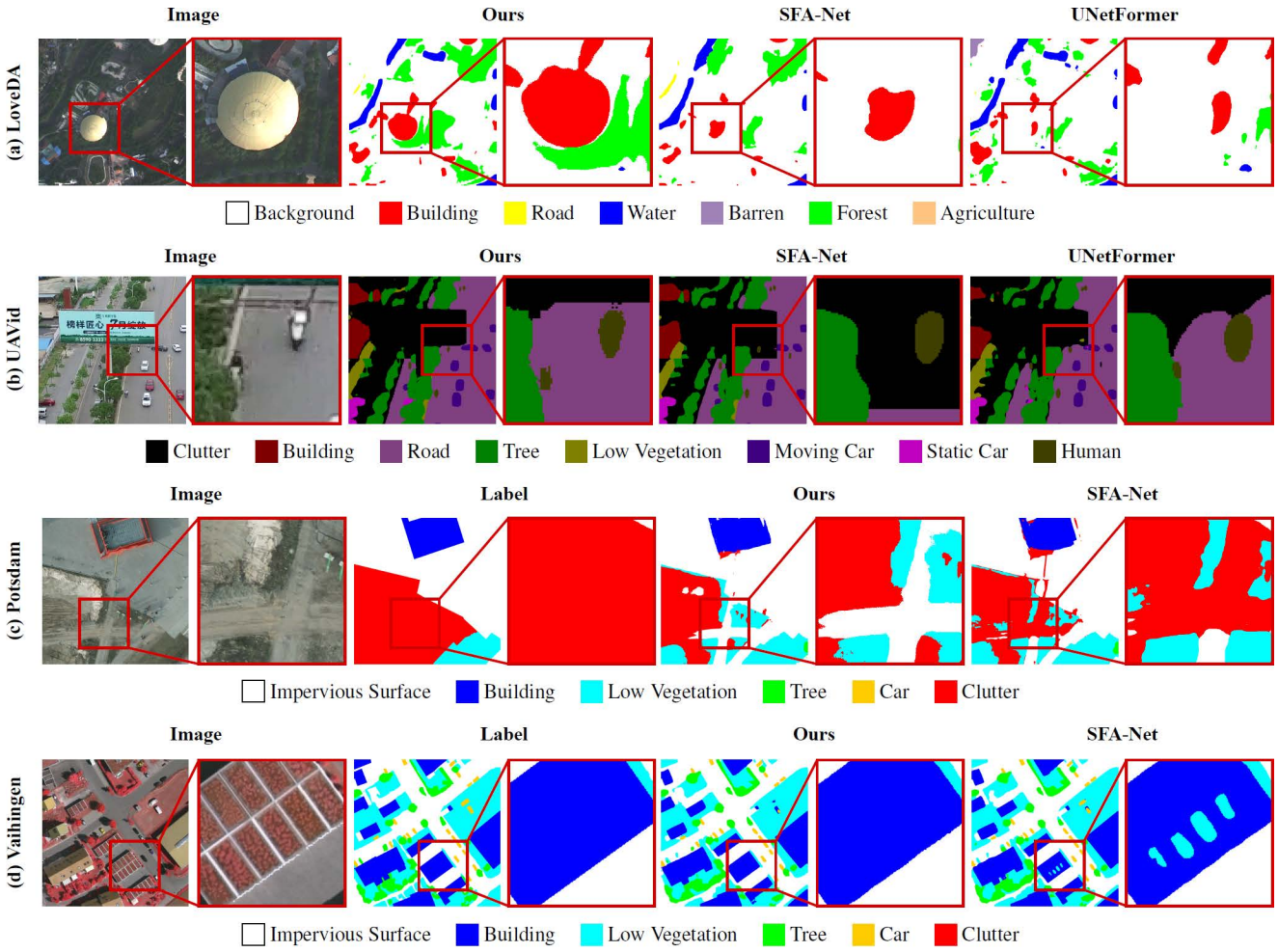


Figure 4. Visualized results on four coarse-grained datasets. Ground truth labels are unavailable in the LoveDA and UAVid datasets.

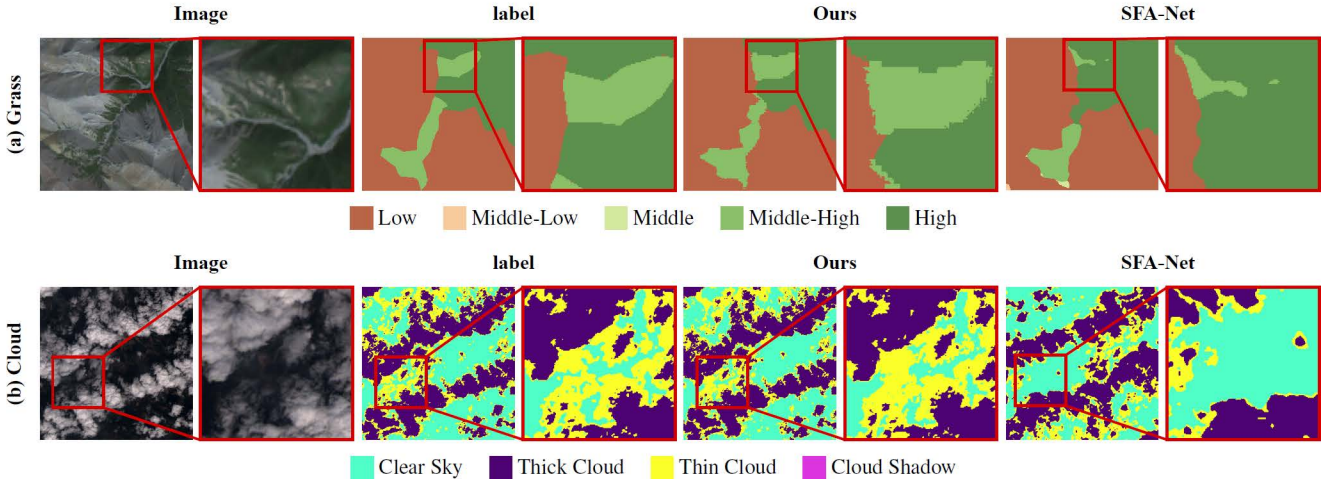


Figure 5. Visualized results on two fine-grained datasets. More visualization of comparative methods can be found in Appendix E.

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